DATA CLEANING DOCUMENTATION

Finova Financial Group: Data Breach Impact Analysis | Prepared by Pearl Pod | October 2026

1. Objective

This document records every check and fix applied to the Finova dataset before analysis. Its purposes are to:

- make sure the fraud, risk and financial figures shown in the CEO, CRO and CFO dashboards can be trusted.
- show the reasoning behind each cleaning decision, including what was kept and why.

2. Data Source

The data is the **DBT Case 002 dataset** supplied by the Databeauty Tribe Business Challenge (BCS-002) for Finova Financial Group. It is a synthetic dataset and it covers January 2025 to August 2026 and follows a single-fact-table star schema.

Table	Role	Columns	Rows
Fact_Transactions	Fact table	TransactionID, DateKey, CustomerKey, AccountKey, BranchKey, ChannelKey, TransactionTypeKey, Amount, Status, SuspiciousFlag, FraudFlag, FeeIncome	11,979
Dim_Customer	Dimension	CustomerKey, CustomerName, Segment, Status, DataExposed, Age	800
Dim_CustomerSegment	Dimension	SegmentKey, Segment (Corporate, SME, Retail, Premium)	4
Dim_Account	Dimension	AccountKey, AccountType, Status (Active, Dormant, Closed), Balance	1,000
Dim_Branch	Dimension	BranchKey, RegionKey	30
Dim_Region	Dimension	RegionKey, Region, State	10
Dim_Channel	Dimension	ChannelKey, Channel	6
Dim_TransactionType	Dimension	TransactionTypeKey, TransactionType	6
Dim_Date	Dimension	DateKey, Year, Quarter, Month (plus MonthNumber, YearMonth)	608
Targets	Monthly targets	Month, RevenueTarget, ProfitTarget, TransactionTarget, MaxFraudCases	20

3. Tools Used

Tool	How it was used
Power BI Desktop: Power Query	Row counts, duplicate detection, column quality, Trim and Clean, data types, null rows, orphan-keys, Add columns (AgeGroup and AgeSort).
Power BI Desktop: DAX	IQR outlier test after Close and Apply.

4. Cleaning steps

4.1 Summary of all steps

Step	Check	Finding	Action	Result
1	Row counts and totals	Baseline row counts, amount and fee totals recorded	None (baseline)	Reference figures set
2	Duplicates	5 identical duplicate rows (same TransactionID)	Removed	12,005 to 12,000 rows
3	Data types	Keys and numbers needed correct types	Set types	All columns typed correctly
4	Trim and Clean	Leading or trailing spaces	Trimmed and cleaned text columns	Text columns consistent
5	Nulls and blanks	21 null Amount rows, all flags = No	Deleted	12,000 to 11,979 rows
6	Date range	Jan 2025 to 31 Aug 2026	None needed	Dates valid
7	Orphan keys	No fact key without a dimension match	None needed	Relationships safe
8	Flag logic	Only Yes and No in both flags	None needed	Flags valid
9	Outliers	899 transactions by IQR test (about 7.5%)	Recorded and kept	No data lost
10	Targets	20 months, no nulls	None needed	Targets clean
11	Derived fields	Age needed grouping; month labels needed.	Created AgeGroup, MonthNumber, YearMonth	Six age brackets added.
12	Final validation	Headline figures cross-checked in both tools	Reconciled	Figures match

4.2 Detailed record of each step

Step 1: Quick validation checks (row counts and totals)

Table used	All tables
Issue checked	Baseline row counts and value totals, recorded before and after cleaning so every change can be traced.
What was found	The fact table started with 12,005 rows and the Amount and FeeIncome totals changed as rows were removed.
Action taken	Recorded the counts at three points: raw, after duplicate removal, and final.
Reason	A before-and-after record proves that each fix did only what it was meant to do.
Result	Reference figures established.

S/N	Table	Check	Raw	After duplicates removed	Final (after null rows removed)
1	Fact_Transactions	Rows	12,005	12,000	11,979
2	Fact_Transactions	Unique transactions	12,000	12,000	11,979
3	Fact_Transactions	Total Amount	₦872,645,481.08	₦872,333,834.55	₦872,333,834.55
4	Fact_Transactions	Total FeeIncome	₦8,368,104.84	₦8,364,983.63	₦8,357,067.72
5	Dim_Customer	Customers	800	800	800
6	Dim_Account	Accounts	1,000	1,000	1,000
7	Dim_Branch	Branches	30	30	30
8	Dim_Region	Rows	10	10	10
9	Dim_Channel	Channels	6	6	6
10	Dim_TransactionType	Transaction types	6	6	6
11	Dim_CustomerSegment	Segments	4	4	4
12	Dim_Date	Dates	608	608	608
13	Targets	Months	20	20	20

The Amount total did not change when the null rows were removed because a null contributes nothing to a sum. The FeeIncome total fell by ₦7,915.91 (about 0.09%) because those rows did carry a small fee amount.

Step 2: Duplicate transactions

Table used	Fact_Transactions
Issue checked	Identical records repeated across all columns, including TransactionID.
What was found	5 duplicate records (T000100, T000201, T000302, T000403, T000504). Keep Rows > Keep Duplicates in Power Query.
Action taken	Removed the duplicates with Remove Duplicates on TransactionID (Power Query)
Reason	A duplicate counts the same transaction twice and inflates revenue, value and fraud numbers.
Result	Fact table reduced from 12,005 to 12,000 rows; Amount total fell from ₦872.65M to ₦872.33M.

Step 3: Data types

Table used	All tables
Issue checked	Whether each column had the correct data type.
What was found	Numeric columns needed proper numeric types, and keys needed to stay as text.
Action taken	Text for all IDs and keys; whole number for Age, MonthNumber and Year; fixed decimal number for Amount, FeeIncome and Balance; date for date columns.
Reason	Correct types allow accurate calculations, sorting and relationships. IDs are labels, not numbers, so they stay as text.
Result	All columns have the right data types

Step 4: Trimming spaces

Table used	All tables
Issue checked	Extra spaces and hidden characters in text columns.
What was found	Leading or trailing spaces.
Action taken	Selected all text columns in all tables and applied Transform > Format > Trim and Clean (Power Query).
Reason	Hidden spaces make values such as "Yes" and "Yes " count as two different values, which breaks grouping and joins.
Result	Text columns are consistent.

Step 5: Nulls and blanks

Table used	Fact_Transactions
Issue checked	Empty cells in every column.
What was found	Column quality was checked for every column. Only Amount had empty cells: 21 rows (0.18% of 12,000). All 21 had FraudFlag = No and SuspiciousFlag = No.
Action taken	The 21 rows were deleted because they are negligible, weren't flagged and carry no amount to analyse.
Reason	A row with no amount cannot contribute to any value measure, so removing it keeps the dataset clean at almost no cost.
Result	No null Amount rows remain.

Step 6: Date range

Table used	Fact_Transactions and Dim_Date
Issue checked	Whether all dates fall within the expected period.
What was found	Earliest date: 1st January 2025. Latest date: 31st August 2026. This matches the case period (Dim_Date has 608 days).
Action taken	None needed. Two methods gave the same answer: column profile statistics, and a reference query with Transform > Date > Earliest and Latest.
Reason	Dates outside the case period would give wrong monthly trends.
Result	All dates are within January 2025 to August 2026.

Step 7: Orphan keys

Table used	Fact_Transactions and each dimension table
Issue checked	Fact keys with no matching row in a dimension table.
What was found	Merge Queries as New with a Left Anti join (fact key to dimension key) returned no rows for any key. The same test between linked dimensions also returned no rows.
Action taken	None needed.
Reason	Orphan keys break relationships and drop records from reports.
Result	No orphan keys.

Step 8: Flag logic

Table used	Fact_Transactions (SuspiciousFlag, FraudFlag)
Issue checked	Whether the flags contain only valid values.
What was found	Both columns contain only Yes and No. In the final data there are 325 suspicious and 71 fraud transactions, and every fraud transaction is also flagged suspicious.
Action taken	None needed.
Reason	The case separates suspicious (not confirmed fraud) from confirmed fraud, so both flags must be clean.
Result	Flags are valid and ready for analysis.

Step 9: Outliers

Table used	Fact_Transactions (Amount)
Issue checked	Unusually high, low, zero or negative amounts.
What was found	Column profile statistics showed the minimum, maximum and average, and there were no zero or negative values. Sorting the Amount column in both directions confirmed it. The IQR method (1.5 × IQR), run in DAX found **899 outliers, about 7.5%** of transactions.
Action taken	All records were retained.
Reason	In fraud analysis, very large transactions may be the signal, so deleting them could hide fraud. Large transactions are also normal in banking.
Result	899 outliers documented and kept in the analysis.

Step 10: Targets table

Table used	Targets
Issue checked	Nulls, data type and date range.
What was found	No nulls. The Month column is a date type and runs from January 2025 to August 2026 (20 months).
Action taken	None needed.
Reason	Targets must match the fact table's months to compare actual against target.
Result	Targets table is clean.

Step 11: Derived fields

Table used	Dim_Customer and Dim_Date
Issue checked	Age is stored as a single number per customer, which is too detailed to compare fraud patterns across groups. Month labels were also needed for monthly trends.
Action taken	Created AgeGroup column with six bands: 18-24, 25-34, 35-44, 45-54, 55-64 and 65+ (the oldest band is open-ended). Added a sort-order column (1 to 6) and used Sort by column so the bands appear in age order instead of alphabetically. Created MonthNumber and YearMonth in Dim_Date for monthly trends.
Reason	Grouping ages into brackets gives enough transactions in each group to compare fraud and suspicious rates, and the brackets follow common banking and marketing age ranges.
Result	Six age brackets created. Among the 584 customers who transacted, the groups are reasonably balanced (85 to 118 customers each).

5. Final Validation

Measure	Final value
Transactions (unique, clean)	11,979
Total transaction value	₦872.33M
Total fee revenue	₦8.36M
Fraud transactions	71 (0.59%)
Suspicious transactions	325 (2.71%)
Fraud amount	₦4.63M (0.53% of transaction value)
Customers	800 in total; 584 made at least one transaction (216 had none in the period)
Exposed customers (DataExposed = Yes)	398 (49.75% of 800)
Accounts	1,000
Period	January 2025 to August 2026 (20 months)

6. Conclusion

The Finova dataset was checked and cleaned in 11 steps before analysis. Five duplicate transactions and 21 null amount rows were removed, leaving 11,979 clean transactions. Dates, keys, flags, and targets passed every validation check, and the 899 outliers were kept because large transactions may be genuine fraud signals. AgeGroup, MonthNumber and YearMonth were added to support age and trend analysis. The final figures reconcile across cleaning and analysis, so the dashboards and report can be reliable.